

Evaluating the Generalization of Machine Learning-Based Network Intrusion Detection Under Temporal, Domain, and Attack-Family Distribution Shifts

Abstract

Machine learning-based Network Intrusion Detection Systems (NIDS) frequently report very high classification performance when evaluated using conventional random train/test splits. However, such evaluation can obscure the effect of temporal variation, domain differences, and changes in attack characteristics that occur in realistic network environments. This study investigates the generalization behavior of a Random Forest-based network intrusion detection model using the CIC-IDS2017 dataset.

The study was designed as a sequence of controlled experiments covering multiclass classification, cross-day generalization, DDoS generalization, false-negative analysis, classification-threshold selection, feature engineering, target-domain exposure, unseen attack-family testing, cross-family DDoS testing, and feature-distribution shift analysis. The baseline multiclass experiment achieved 99.90% accuracy, precision, recall, and F1-score under a stratified random split. However, substantially lower performance was observed when the evaluation protocol was changed to more realistic distribution-shift settings. Cross-day generalization produced an F1-score of 0.4382, while DDoS generalization achieved an F1-score of 0.7764 with an attack recall of approximately 63.5%.

The false-negative analysis showed that 36,493 of 100,000 DDoS samples were missed in the principal DDoS generalization experiment. Threshold analysis showed that reducing the decision threshold did not solve the fundamental recall limitation without increasing false positives. Feature engineering produced only marginal changes in performance. In contrast, controlled exposure to target-domain DDoS samples increased performance dramatically, reaching an F1-score of 0.9993 when 50% target-domain exposure was used. Cross-attack-family experiments further demonstrated that models trained on one attack family frequently failed to detect attacks belonging to another family. Distribution-shift analysis confirmed substantial statistical differences between DDoS, DoS, and Patator traffic.

The results demonstrate that extremely high performance under random splitting should not be interpreted as evidence of real-world generalization. For practical NIDS evaluation, temporal separation, attack-family separation, target-domain testing, false-negative analysis, and feature-distribution analysis are essential.

Keywords: Network Intrusion Detection, Machine Learning, Random Forest, CIC-IDS2017, DDoS Detection, Generalization, Distribution Shift, Concept Drift, Cross-Day Evaluation, Attack-Family Generalization

1. Introduction

Network intrusion detection systems are an important component of modern cybersecurity infrastructure. Their primary purpose is to identify malicious or anomalous network activity and distinguish it from legitimate traffic. Traditional intrusion detection approaches rely heavily on predefined signatures or manually engineered rules. Although signature-based systems remain useful for known threats, they are less effective when attack characteristics change or previously unseen attack behavior is encountered.

Machine learning provides an alternative approach by learning statistical patterns from network traffic. Supervised learning algorithms can classify network flows according to features such as packet lengths, packet rates, inter-arrival times, header information, flow duration, and transport-layer characteristics. Among conventional machine learning algorithms, Random Forest is particularly attractive for intrusion detection because it can model nonlinear relationships, handle heterogeneous features, and provide feature-importance estimates.

A major problem, however, is the evaluation methodology used in many machine learning intrusion-detection studies. A model can achieve extremely high performance when samples from the same traffic distribution are randomly divided between training and testing sets. This does not necessarily mean that the model will perform well on traffic generated on another day, in another environment, or by a different attack family.

This study therefore focuses not simply on whether a machine learning model can classify CIC-IDS2017 traffic, but on **how well the learned model generalizes when the data distribution changes**.

The research investigates several forms of distribution shift:

1. Random train/test evaluation.
2. Temporal or cross-day generalization.
3. DDoS attack generalization.
4. False-negative behavior.
5. Decision-threshold sensitivity.
6. Feature engineering.
7. Target-domain exposure.
8. Unseen attack-family transfer.
9. Cross-family DDoS testing.
10. Statistical feature-distribution shift.

The central research question is:

To what extent does a machine learning-based NIDS trained on CIC-IDS2017 generalize when the distribution of network traffic changes across time, domains, and attack families?

The study further asks:

- Does conventional random splitting overestimate NIDS performance?
- How large is the performance degradation under cross-day evaluation?

- Which characteristics distinguish detected and missed DDoS attacks?
 - Can decision-threshold adjustment compensate for poor generalization?
 - Does feature engineering improve robustness?
 - How much target-domain data is required to recover performance?
 - Can a model trained on one attack family detect another family?
 - Are poor generalization results associated with measurable feature-distribution differences?
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2. Related Work

Machine learning has been extensively investigated for network intrusion detection using benchmark datasets such as CIC-IDS2017. These datasets contain network flows representing benign traffic and multiple categories of attacks, including denial-of-service attacks, distributed denial-of-service attacks, port scanning, brute-force attacks, bot activity, infiltration, and web attacks.

A recurring methodological issue in intrusion-detection research is the use of random train/test partitions. Random partitioning is useful for estimating performance when future samples are assumed to come from the same distribution as the training data. However, network traffic is inherently dynamic. Traffic patterns can vary with time, users, applications, network configurations, operating systems, attack tools, and attack implementations.

Consequently, high accuracy under a random split does not necessarily establish robustness to distribution shift.

Temporal evaluation provides a more realistic alternative by training on traffic from one period and testing on traffic from another. Similarly, attack-family evaluation can determine whether models learn general characteristics of malicious traffic or instead memorize characteristics associated with particular attacks.

Another important issue is class imbalance. In intrusion datasets, some attacks may contain hundreds of thousands of flows while others contain only a few dozen. Overall accuracy can therefore hide poor detection of minority or difficult attack classes. Precision, recall, F1-score, and confusion matrices provide more informative measures.

This study extends these evaluation ideas by combining temporal evaluation, attack-family transfer, target-domain exposure, false-negative analysis, threshold analysis, and feature-distribution testing within a single experimental framework.

3. Dataset

3.1 CIC-IDS2017

The experiments use the CIC-IDS2017 network intrusion dataset. The available dataset files contain traffic collected across multiple days and include both benign traffic and several attack categories.

The dataset inventory identified **15 unique labels**:

- BENIGN
- Bot
- DDoS
- DoS GoldenEye
- DoS Hulk
- DoS Slowhttptest
- DoS slowloris
- FTP-Patator
- Heartbleed
- Infiltration
- PortScan
- SSH-Patator
- Web Attack – Brute Force
- Web Attack – SQL Injection
- Web Attack – XSS

The original files used during preprocessing included Monday through Friday traffic and contained approximately one million records per day in total when combined across the available files.

The dataset was inspected before model development to identify class distributions and determine which attack categories were sufficiently represented for controlled experiments.

3.2 Dataset Inventory

The inventory revealed substantial differences in class frequency.

Monday contained only BENIGN traffic. Tuesday contained BENIGN, FTP-Patator, and SSH-Patator traffic. Wednesday contained BENIGN traffic and several DoS categories. Thursday contained web attacks and infiltration traffic. Friday contained Bot, PortScan, and DDoS traffic.

This structure is particularly important for the research because the attack categories are not uniformly distributed across time. Consequently, temporal evaluation naturally introduces distribution differences between training and testing data.

4. Methodology

4.1 Data Preprocessing

The network-flow CSV files were combined and processed for supervised classification. Non-informative and problematic values were handled during preprocessing, including infinite numerical values and missing values.

The primary classifier used throughout the experiments was a **Random Forest** model.

Random Forest was selected because it:

- handles nonlinear relationships;
- can model interactions between network-flow features;
- requires relatively limited feature scaling;
- provides feature-importance estimates;
- performs well on structured tabular data.

A fixed random state of 42 was used for reproducibility in the principal experiments.

4.2 Baseline Binary Classification

The initial experiment treated network traffic as a binary classification problem:

- BENIGN
- ATTACK

The baseline model produced:

Metric	Result
Accuracy	0.9999
Precision	1.0000
Recall	0.9999
F1-score	0.9999
ROC-AUC	1.0000

The confusion matrix contained:

	Predicted BENIGN	Predicted ATTACK
Actual BENIGN	19,014	0
Actual ATTACK	3	25,600

Only three attack samples were incorrectly classified as benign.

A second binary experiment removed **Destination Port** from the feature set. Performance remained almost unchanged, demonstrating that the initial binary classification result was not solely dependent on the destination-port feature.

5. Experimental Design

The experiments were organized into two major stages.

5.1 Experiment 3: Generalization Analysis

Experiment 3 investigated whether the strong random-split performance remained stable under more challenging evaluation conditions.

Experiment 3A: Multiclass Classification

Ten classes were selected for multiclass classification:

1. BENIGN
2. Bot
3. DDoS
4. DoS GoldenEye
5. DoS Hulk
6. DoS Slowhttptest
7. DoS slowloris
8. FTP-Patator
9. PortScan
10. SSH-Patator

A maximum of 100,000 samples per class was used, with an 80/20 stratified train/test split.

The model achieved:

- Accuracy: **0.9990**
- Precision: **0.9990**
- Recall: **0.9990**
- F1-score: **0.9990**

The weakest class was Bot, with an F1-score of approximately 0.97. Most other classes achieved approximately 1.00 F1-score.

The result demonstrates that conventional random splitting produces extremely strong performance on this dataset.

However, this result alone cannot establish generalization to future or unseen traffic.

5.2 Experiment 3B: Cross-Day Generalization

Experiment 3B changed the evaluation methodology by separating training and testing traffic according to temporal or day-specific characteristics.

The model achieved:

Metric	Result
Accuracy	0.6398
Precision	0.9952
Recall	0.2810
F1-score	0.4382

The large reduction from the 0.9990 random-split F1-score demonstrates that the model's performance is highly sensitive to changes in traffic distribution.

The most important observation is the difference between precision and recall. Precision remained extremely high, but recall fell to only 28.1%. This means that when the model predicted an attack, it was usually correct, but it failed to identify a large proportion of actual attacks.

This is particularly important for intrusion detection because missed attacks can be more operationally significant than false alarms.

5.3 Experiment 3C: DDoS Generalization

Experiment 3C focused specifically on DDoS detection under a generalization setting.

The model achieved:

Metric	Result
Accuracy	0.8149
Precision	0.9986
Recall	0.6351
F1-score	0.7764

The result indicates that DDoS detection was substantially better than the broader cross-day experiment but remained far from perfect.

The precision of 0.9986 shows that the model generated very few false alarms. However, recall of 0.6351 means that approximately 36.5% of the DDoS attacks were missed.

6. Experiment 3D: DDoS False-Negative Analysis

Because recall was substantially lower than precision, Experiment 3D investigated the characteristics of missed DDoS flows.

Out of 100,000 DDoS attacks:

- Correctly detected: **63,507**
- False negatives: **36,493**
- Recall: **0.6351**
- False-negative rate: **0.3649**

The analysis compared feature statistics for detected and missed attacks.

Several features showed substantial differences, including:

- URG Flag Count
- Min Packet Length
- Fwd Packet Length Min
- Flow IAT Min
- Bwd Packet Length Min
- Fwd IAT Min
- Fwd Packets/s
- Flow Packets/s
- Bwd Packets/s
- ACK Flag Count

For example, Flow IAT Min showed a large difference between detected and missed attacks, while packet-length-related features also differed considerably.

These results suggest that the false negatives were not simply random classification errors. They were associated with substantially different network-flow characteristics.

This supports the hypothesis that the model learned patterns strongly associated with particular manifestations of DDoS traffic and had difficulty recognizing traffic exhibiting different characteristics.

7. Experiment 4: Robustness and Distribution Shift

Experiment 4 investigated possible methods for improving or explaining the generalization problem identified in Experiment 3.

7.1 Experiment 4A: Classification Threshold Analysis

The default classification threshold was varied from 0.90 to 0.10.

The principal results were:

Threshold	Accuracy	Precision	Recall	F1
0.90	0.4940	0.0000	0.0000	0.0000
0.80	0.5036	0.9815	0.0191	0.0375
0.70	0.6578	0.9985	0.3241	0.4893
0.60	0.8150	0.9991	0.6349	0.7764
0.50	0.8149	0.9986	0.6351	0.7764
0.40	0.8149	0.9981	0.6353	0.7764
0.30	0.8146	0.9970	0.6355	0.7762
0.20	0.8093	0.9793	0.6364	0.7715
0.10	0.7810	0.8993	0.6386	0.7468

The best F1-score occurred at a threshold of 0.60:

F1 = 0.7764

The threshold analysis demonstrates that changing the decision threshold could slightly alter the precision-recall balance, but it could not eliminate the fundamental recall limitation.

Even at a threshold of 0.10, recall increased only to 0.6386 while precision decreased to 0.8993.

Therefore, the principal limitation was not simply an inappropriate classification threshold.

8. Experiment 4B: Feature Engineering

Experiment 4B introduced engineered network-flow features designed to express relationships between traffic volume, packet counts, bytes, and flow duration.

Examples included:

- Bwd_Packets_Per_Duration
- Fwd_Bwd_Packet_Length_Ratio
- Log_Bwd_Packets_per_s

- Fwd_Byte_Proportion
- Bwd_Bytes_Per_Packet
- Fwd_Bwd_Byte_Ratio
- Bwd_Fwd_Byte_Ratio
- Fwd_Bytes_Per_Duration
- Bwd_Byte_Proportion
- Log_Packet_Length_Mean

The resulting performance was:

Metric	Result
Accuracy	0.8147
Precision	0.9986
Recall	0.6346
F1-score	0.7760

The improvement was negligible compared with Experiment 3C.

This is an important negative finding. Simply creating additional ratios and transformations did not solve the generalization problem.

The most important engineered feature was Bwd_Packets_Per_Duration, followed by Fwd_Bwd_Packet_Length_Ratio and Log_Bwd_Packets_per_s.

9. Experiment 4C: Target-Domain Exposure

Experiment 4C investigated whether limited exposure to target-domain DDoS traffic could improve generalization.

The amount of target-domain training data was varied.

Target Exposure	Target Samples	Accuracy	Precision	Recall	F1
0%	0	0.7927	0.9987	0.6352	0.7765
5%	3,200	0.9985	0.9992	0.9981	0.9986
10%	6,401	0.9988	0.9992	0.9986	0.9989
25%	16,003	0.9991	0.9992	0.9993	0.9992
50%	32,007	0.9992	0.9992	0.9995	0.9993

The result is striking.

With no target-domain exposure, F1-score was approximately 0.7765. With only 5% target-domain exposure, F1-score increased to approximately 0.9986.

At 50% exposure, F1-score reached approximately 0.9993.

This experiment provides strong evidence that the observed performance degradation is closely related to distribution differences between the training and target data.

Importantly, the result should not be interpreted as evidence that the model inherently generalizes well. Instead, it demonstrates that **target-domain adaptation or exposure can substantially recover performance**.

10. Experiment 4D: Unseen Attack-Family Testing

Experiment 4D evaluated the model against an attack family that was not represented in the training distribution in the same way.

The result was:

Metric	Result
Accuracy	0.8710
Precision	0.0035
Recall	0.0002
F1-score	0.0004

The confusion matrix showed:

- False positives: 853
- False negatives: 13,832
- True positives: 3
- True negatives: 99,147

Although overall accuracy was 87.1%, the attack-detection performance was essentially unusable.

Only **3 of 13,835 attacks** were correctly identified.

This experiment provides one of the clearest demonstrations in the study that accuracy can be misleading in intrusion detection.

A model can report relatively high accuracy while almost completely failing to detect the target attack class.

11. Experiment 4E: Cross-Family DDoS Testing

Experiment 4E evaluated cross-family behavior using DDoS-related data.

The model produced:

Metric	Result
Accuracy	0.8143
Precision	0.9989
Recall	0.6335
F1-score	0.7753

The confusion matrix contained:

- False positives: 69
- False negatives: 36,651
- True positives: 63,349
- True negatives: 97,649

Again, precision was extremely high while recall remained around 63%.

This reinforces the pattern observed in Experiment 3C: the model is conservative in identifying attacks. When it raises an alarm it is generally correct, but a substantial proportion of attacks remain undetected.

12. Experiment 4F: Attack-Family Generalization

Experiment 4F directly examined transfer between three attack families:

- DDoS
- DoS
- Patator

The results were:

Training Family	Testing Family	Accuracy	Precision	Recall	F1
DDoS	DoS	0.4999	0.3846	0.0004	0.0008
DDoS	Patator	0.7833	0.0000	0.0000	0.0000
DoS	DDoS	0.8169	0.9984	0.6348	0.7761
DoS	Patator	0.7747	0.0055	0.0002	0.0004

Training Family	Testing Family	Accuracy	Precision	Recall	F1
Patator	DDoS	0.5000	0.0000	0.0000	0.0000
Patator	DoS	0.5001	0.8182	0.0004	0.0007

The transfer behavior was highly asymmetric.

For example, a model trained on DDoS traffic failed almost completely when evaluated on DoS traffic. Similarly, training on Patator traffic provided almost no useful detection of DDoS or DoS attacks.

These findings indicate that the model does not simply learn a universal representation of malicious network activity. Instead, it appears to learn traffic characteristics strongly associated with the training attack family.

13. Experiment 4G: Feature Distribution Shift

To investigate why attack-family transfer failed, Experiment 4G measured statistical differences between feature distributions using the Kolmogorov-Smirnov statistic.

Large KS values indicate substantial differences between two distributions.

The strongest observed shifts included:

Attack Families	Feature	KS Statistic
DoS vs Patator	Bwd Packets/s	0.7978
DoS vs DDoS	Init_Win_bytes_backward	0.6916
DoS vs DDoS	Total Length of Fwd Packets	0.6568
DoS vs DDoS	Fwd Packet Length Max	0.6568
DoS vs Patator	Average Packet Size	0.6515
DoS vs Patator	Packet Length Mean	0.6504
DoS vs Patator	Packet Length Std	0.6416
DDoS vs Patator	Average Packet Size	0.6318

The pairwise distribution-shift statistics were:

Pair	Mean KS	Median KS	Maximum KS
DDoS vs Patator	0.4871	0.4982	0.6318

Pair	Mean KS	Median KS	Maximum KS
DoS vs DDoS	0.4353	0.4230	0.6916
DoS vs Patator	0.5221	0.6273	0.7978

These are substantial distribution differences.

The results provide quantitative support for the hypothesis that cross-family generalization fails partly because the statistical characteristics of the network flows differ considerably between attack families.

14. Feature Importance Analysis

Feature importance analysis was conducted throughout the experiments to identify the network-flow characteristics most heavily used by the Random Forest model.

For the multiclass model, Destination Port was the most important feature, with an importance of approximately 0.0684.

Other highly ranked features included:

- Bwd Header Length
- Packet Length Mean
- Init_Win_bytes_forward
- Flow IAT Mean
- Average Packet Size
- Fwd Packet Length Max
- Init_Win_bytes_backward
- Subflow Fwd Bytes
- Flow IAT Max
- Bwd Packet Length Max

The binary baseline also identified features such as Init_Win_bytes_forward, Fwd Packet Length Max, Bwd Packet Length Max, Fwd Packet Length Mean, and packet-segment statistics as important.

After feature engineering, Bwd_Packets_Per_Duration became the most important engineered feature.

These results suggest that the classifier relies heavily on packet size, packet timing, packet-rate, TCP-window, and forward/backward traffic relationships.

However, high feature importance should not be interpreted as evidence that these features are universally reliable indicators of malicious traffic. Experiment 4G demonstrates that several of these feature distributions change substantially between attack families.

15. Results Summary

The major experimental results are summarized below.

Experiment	Purpose	Accuracy	Precision	Recall	F1
3A	Multiclass classification	0.9990	0.9990	0.9990	0.9990
3B	Cross-day generalization	0.6398	0.9952	0.2810	0.4382
3C	DDoS generalization	0.8149	0.9986	0.6351	0.7764
3D	DDoS false-negative analysis	—	—	0.6351	—
4A	Threshold analysis	0.8150	0.9991	0.6349	0.7764
4B	Feature engineering	0.8147	0.9986	0.6346	0.7760
4C	Target-domain exposure	0.9992	0.9992	0.9995	0.9993
4D	Unseen attack-family testing	0.8710	0.0035	0.0002	0.0004
4E	Cross-family DDoS testing	0.8143	0.9989	0.6335	0.7753

Experiments 4F and 4G are primarily transfer and distribution analyses rather than conventional single-test-set evaluations and are therefore best interpreted through their family-transfer and KS-statistic results.

16. Discussion

16.1 Random-Split Performance Is Not Sufficient

The most important finding is the difference between Experiment 3A and Experiment 3B.

The model achieved an F1-score of 0.9990 under random stratified evaluation but only 0.4382 under cross-day generalization.

This represents a dramatic reduction in performance.

The finding suggests that random splitting allows training and testing samples to share highly similar statistical characteristics. Consequently, the resulting performance may represent interpolation within a known distribution rather than genuine generalization to future traffic.

For intrusion detection, this distinction is critical.

16.2 Precision and Recall Tell Different Stories

Across several experiments, precision remained close to 1.0 while recall was substantially lower.

For DDoS generalization:

- Precision = 0.9986
- Recall = 0.6351

This means the model was highly conservative. Its alerts were usually correct, but it failed to detect a substantial fraction of attacks.

An operational NIDS cannot be evaluated solely by asking how many alerts are correct. It must also answer how many attacks are missed.

Therefore, recall and false-negative rate should receive particular attention in future evaluations.

16.3 Threshold Adjustment Is Not a Complete Solution

Experiment 4A showed that reducing the classification threshold could slightly improve recall, but only at the cost of increasing false positives.

The best F1-score occurred at a threshold of 0.60, but the resulting recall remained approximately 0.635.

Therefore, the problem is not simply that the classifier's decision boundary is too conservative.

The underlying feature distributions of detected and missed attacks are different. This indicates a representation or distribution problem rather than merely a threshold problem.

16.4 Feature Engineering Alone Is Insufficient

Experiment 4B produced almost the same results as Experiment 3C.

This suggests that adding manually constructed ratios and logarithmic transformations does not automatically provide robustness against distribution shift.

The model still performed well when traffic resembled its training distribution and poorly when traffic differed.

This is an important negative result because it prevents an overly simplistic conclusion that additional feature engineering is sufficient to solve the problem.

16.5 Target-Domain Exposure Is Highly Effective

Experiment 4C produced one of the strongest findings in the study.

Adding only 5% target-domain samples increased the F1-score from approximately 0.7765 to 0.9986.

This demonstrates that even limited knowledge of the target distribution can substantially improve model performance.

However, this also reinforces the distribution-shift explanation.

If a small amount of target-domain data produces such a large improvement, the original model was not failing because the classification task was inherently impossible. It was failing because the learned decision structure did not transfer effectively to the new distribution.

16.6 Attack-Family Transfer Is Difficult

Experiment 4F showed that attack-family transfer frequently failed.

A model trained on one family could perform very poorly when exposed to another family.

This suggests that the classifier learns family-specific traffic signatures rather than general maliciousness.

For example, DDoS traffic can contain high packet rates and distinctive packet-size patterns, while Patator traffic can exhibit different timing and connection characteristics.

A model that relies heavily on such characteristics may perform well within the original family but fail when those characteristics disappear.

16.7 Distribution Shift Explains Much of the Transfer Failure

Experiment 4G provides statistical evidence supporting this interpretation.

Mean KS statistics ranged from approximately 0.435 to 0.522 across the examined attack-family pairs, with maximum values reaching approximately 0.798.

These are substantial distribution differences.

Therefore, the poor cross-family performance is consistent with measurable changes in the underlying feature distributions.

17. Research Findings

The study produces six principal findings.

Finding 1: Conventional evaluation substantially overestimates apparent performance.

Random stratified evaluation produced an F1-score of 0.9990, while cross-day generalization produced only 0.4382.

Finding 2: DDoS detection remains vulnerable to false negatives.

Approximately 36.5% of DDoS attacks were missed under the principal DDoS generalization experiment.

Finding 3: Threshold optimization cannot eliminate distribution-shift problems.

The best threshold produced an F1-score of 0.7764, almost identical to the default-threshold result.

Finding 4: Simple feature engineering provides limited robustness.

Engineered features changed performance only marginally, producing an F1-score of 0.7760.

Finding 5: Target-domain exposure dramatically improves performance.

Five percent target-domain exposure increased F1-score to approximately 0.9986, demonstrating the value of domain adaptation or representative target data.

Finding 6: Attack-family transfer is highly unreliable.

Models trained on one attack family frequently failed to identify attacks from another family, while statistical analysis revealed substantial feature-distribution differences between families.

18. Limitations

Several limitations should be acknowledged.

First, all experiments were performed using CIC-IDS2017. Although it is a widely used benchmark dataset, it represents a particular collection environment and cannot fully represent the diversity of operational networks.

Second, the Random Forest algorithm was the primary model used in the experiments. The observed generalization behavior may differ for other algorithms, including gradient boosting, neural networks, support-vector machines, or modern deep-learning architectures.

Third, the study focuses primarily on flow-level features rather than raw packet payloads. Payload-based information may provide additional discriminative characteristics.

Fourth, some attack categories in CIC-IDS2017 contain very small numbers of samples. This limits the ability to perform statistically strong experiments for every class.

Fifth, the target-domain exposure experiment demonstrates that additional target data can dramatically improve performance, but it does not establish the minimum amount of data required in a new operational environment.

Finally, the results are based on offline evaluation. Real-world deployment introduces additional challenges including concept drift, encrypted traffic, changing applications, adaptive adversaries, sensor placement, latency requirements, and class-prior changes.

19. Future Work

Future research should extend this work in several directions.

19.1 Cross-Dataset Evaluation

The strongest next step is evaluation on an entirely different intrusion-detection dataset rather than another subset of CIC-IDS2017.

This would provide a stronger test of whether the learned representation generalizes beyond the original collection environment.

19.2 Online and Continual Learning

Because network traffic evolves continuously, future models should be evaluated using continual-learning or online-learning strategies.

A practical NIDS should be capable of updating its knowledge as traffic patterns change.

19.3 Domain Adaptation

The strong results from Experiment 4C suggest that domain adaptation deserves further investigation.

Future experiments could compare:

- supervised fine-tuning;
- unsupervised domain adaptation;
- transfer learning;
- importance weighting;

- adversarial domain adaptation;
- incremental retraining.

19.4 Robust Feature Representation

Future research should investigate representations that capture general properties of malicious behavior rather than characteristics tied to individual attacks.

For example, temporal patterns, bidirectional flow relationships, connection-level behavior, and multi-flow context could be investigated.

19.5 Explainable False-Negative Detection

The false-negative analysis should be expanded using explainability techniques such as SHAP or permutation-based analysis.

The objective would be to determine why specific attacks are classified as benign rather than merely identifying statistical differences.

19.6 More Realistic Evaluation Protocols

Future evaluations should prioritize:

- temporal holdout;
- cross-network testing;
- cross-dataset testing;
- unseen attack testing;
- attack-family transfer;
- changing class distributions;
- gradual concept drift.

Such protocols would provide a more realistic assessment of NIDS robustness.

20. Conclusion

This study evaluated a Random Forest-based machine learning Network Intrusion Detection System using CIC-IDS2017 under increasingly challenging experimental conditions.

The initial results were extremely strong. The multiclass random-split experiment achieved 99.90% accuracy, precision, recall, and F1-score. However, subsequent experiments demonstrated that this performance does not represent the complete behavior of the model.

When evaluated under cross-day distribution changes, F1-score dropped to 0.4382. DDoS generalization produced an F1-score of 0.7764 but detected only approximately 63.5% of attacks. False-negative analysis

identified substantial differences between detected and missed attacks. Threshold adjustment produced only marginal improvement, while feature engineering did not materially improve robustness.

The most significant improvement occurred when target-domain data were introduced. With only 5% target-domain exposure, F1-score increased to approximately 0.9986. This indicates that distribution alignment has a much stronger influence on performance than threshold tuning or simple feature engineering.

The attack-family experiments provide further evidence. Models trained on one attack family often failed almost completely when evaluated on another family. Statistical analysis using the Kolmogorov-Smirnov statistic demonstrated substantial feature-distribution differences between DDoS, DoS, and Patator traffic.

The overall conclusion is therefore clear:

A machine learning NIDS can achieve near-perfect benchmark performance while still exhibiting substantial weaknesses in realistic generalization scenarios.

Consequently, high accuracy obtained from a conventional random train/test split should not be treated as sufficient evidence of NIDS reliability. Evaluation should explicitly test temporal generalization, attack-family transfer, false-negative behavior, distribution shift, and adaptation to target environments.

The central contribution of this study is not the 99.90% baseline accuracy. Rather, it is the demonstration that **evaluation methodology strongly determines the apparent effectiveness of a machine learning intrusion detection system**, and that distribution shift is a critical factor that must be considered when assessing practical NIDS robustness.

21. Recommended Figures

The following figures generated during the experiments should be incorporated into the final paper.

Figure 1. Generalization performance across experimental settings.

Figure 2. Effect of target-domain exposure on model performance.

Figure 3. Attack-family generalization performance.

Figure 4. Feature-distribution shift between attack families.

Figure 5. DDoS confusion matrix.

Figure 6. Top model feature importance.

Figure 7. Feature differences between detected and missed DDoS attacks.

Figure 8. Classification-threshold analysis.

The figures should be referenced in the Results and Discussion sections rather than placed without explanation.

22. Reproducibility

The experiments were implemented using Python in a virtual environment on Windows.

The project generated the following principal result artifacts:

- `baseline_results.txt`
- `model_results.txt`
- `model_results_no_port.txt`
- `dataset_inventory.csv`
- `experiment3a_results.txt`
- `experiment3b_results.txt`
- `experiment3c_results.txt`
- `experiment3d_results.txt`
- `experiment4a_results.txt`
- `experiment4b_results.txt`
- `experiment4c_domain_adaptation.csv`
- `experiment4d_results.txt`
- `experiment4e_results.txt`
- `experiment4f_results.txt`
- `experiment4g_results.txt`
- `master_experiment_results.csv`
- `master_experiment_results.txt`

The final visual analysis contains eight figures corresponding to the principal generalization, adaptation, attack-family, distribution-shift, confusion-matrix, feature-importance, false-negative, and threshold analyses.

References

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Final Research Statement

The experiments collectively show that **benchmark accuracy and operational robustness are not equivalent**. The model demonstrated excellent performance when training and testing distributions were closely aligned, but performance deteriorated substantially under temporal and attack-family distribution shifts. The ability of limited target-domain exposure to restore performance further demonstrates that distribution alignment is a central factor in machine-learning-based intrusion detection.

This finding should guide both future research and practical evaluation: **a credible NIDS evaluation must test whether a model can detect attacks it has not effectively seen before, under traffic conditions that differ from its training environment.**